SPLUNK PROJECT

Presented by: Sara Hazem

PROJECT OVERVIEW & GENERAL CONCEPT

- PROJECT THEME: CYBERSECURITY INCIDENT RESPONSE AND LOG ANALYSIS USING SPLUNK ENTERPRISE.
- GENERAL CONCEPT: THIS PROJECT SIMULATES A REAL-WORLD SECURITY INCIDENT SCENARIO FOCUSING ON A COMPROMISED WEB SERVER. IT INVESTIGATES A MULTI-VECTOR ATTACK CAMPAIGN WHERE THREAT ACTORS UTILIZED AUTOMATED TOOLS, VULNERABILITY SCANNING, REMOTE COMMAND EXECUTION, AND MALICIOUS FILE STAGING.
- CORE SCOPE: THE PROJECT BRIDGES RAW LOG DATA WITH ACTIONABLE INTELLIGENCE BY EXAMINING WEB TRAFFIC, HTTP USER-AGENTS, REQUEST PATHS, AND FIREWALL TELEMETRY TO UNCOVER THE FULL LIFECYCLE OF A CYBER ATTACK.

PROJECT OBJECTIVES & BUSINESS/SECURITY VALUE

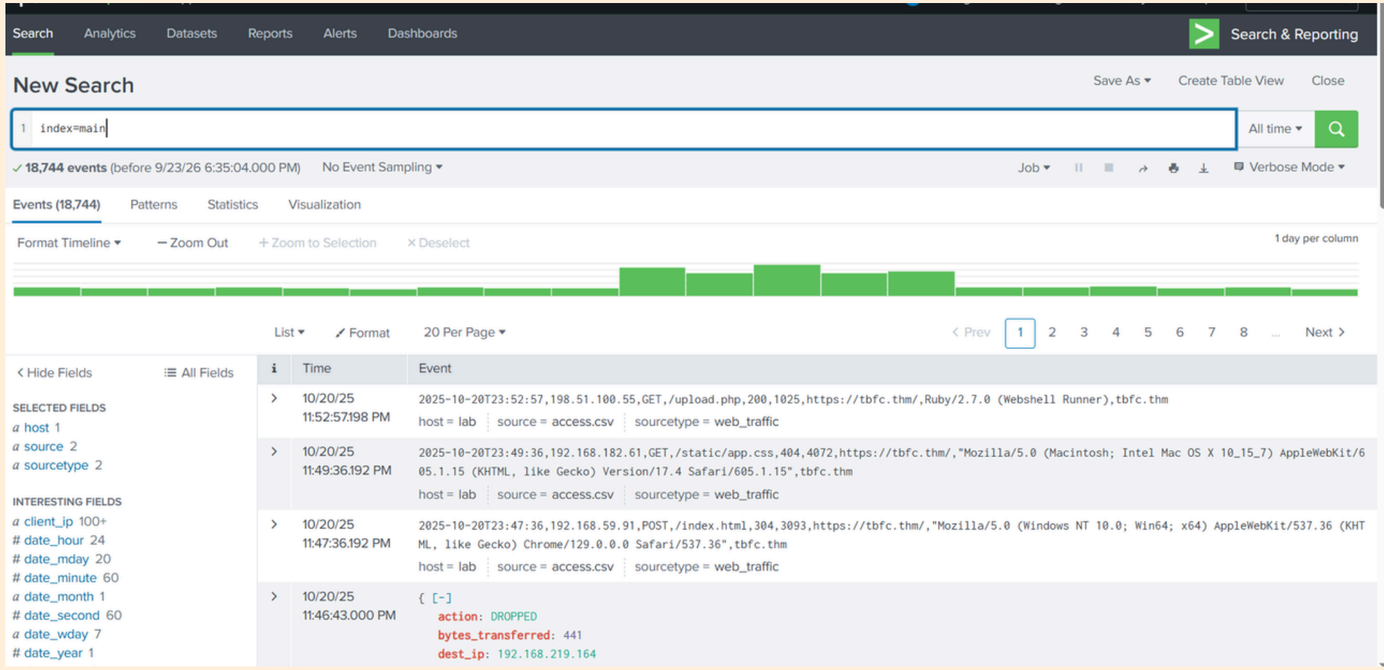
- CORE OBJECTIVES: THE PRIMARY GOAL IS TO EMPOWER SECURITY OPERATIONS CENTER (SOC) ANALYSTS TO SWIFTLY DETECT, SCOPE, AND MITIGATE ADVANCED CYBER THREATS USING INDUSTRY-STANDARD SIEM TOOLS.
- WHY THIS PROJECT MATTERS: IT DEMONSTRATES THE CRITICAL CAPABILITY OF TRANSFORMING MASSIVE VOLUMES OF UNSTRUCTURED LOG DATA INTO CLEAR, CHRONOLOGICAL NARRATIVES OF A SECURITY BREACH.
- PRACTICAL UTILITY: IT PROVIDES HANDS-ON EXPERTISE IN IDENTIFYING HOW ATTACKERS LEVERAGE AUTOMATED SCRIPTS (SUCH AS SQLMAP AND HAVIJ) AND ESTABLISH PERSISTENCE, GIVING DEFENDERS THE BLUEPRINT NEEDED TO HARDEN CORPORATE INFRASTRUCTURES AND PREVENT FUTURE COMPROMISES.

START OF THE PROJECT

Let's

GO!

- WHAT WAS DONE:
- INITIALIZED THE SPLUNK SIEM SEARCH INTERFACE BY QUERYING THE DEFAULT INDEX (INDEX=MAIN) TO INSPECT INGESTED WEB SERVER LOGS (SOURCE=ACCESS.CSV, SOURCETYPE=WEB_TRAFFIC). THIS STEP VERIFIES THAT LOG DATA IS SUCCESSFULLY FLOWING INTO THE PLATFORM, DISPLAYING A TOTAL VOLUME OF OVER 18,000 RAW SECURITY AND WEB ACCESS EVENTS.
- OUTPUTS & FINDINGS:
- THE TIMELINE HISTOGRAM AND EVENT LOGS CONFIRM ACTIVE DATA INGESTION. INITIAL EVENT PREVIEWS DISPLAY SUSPICIOUS ENTRIES, SUCH AS REQUESTS POINTING TO /UPLOAD.PHP ASSOCIATED WITH TOOLS LIKE RUBY/2.7.0 (WEBSHELL RUNNER), INDICATING POTENTIAL POST-EXPLOITATION OR BACKDOOR ACTIVITY.
- SECURITY IMPLICATIONS:
- ESTABLISHING BASELINE VISIBILITY INTO WEB TRAFFIC LOGS IS THE CRUCIAL FIRST STEP IN SECURITY OPERATIONS CENTER (SOC) INVESTIGATIONS, ENABLING ANALYSTS TO SPOT MALICIOUS ARTIFACTS, UNAUTHORIZED ACCESS VECTORS, AND ANOMALOUS TOOL SIGNATURES EARLY IN THE TRIAGE PROCESS.



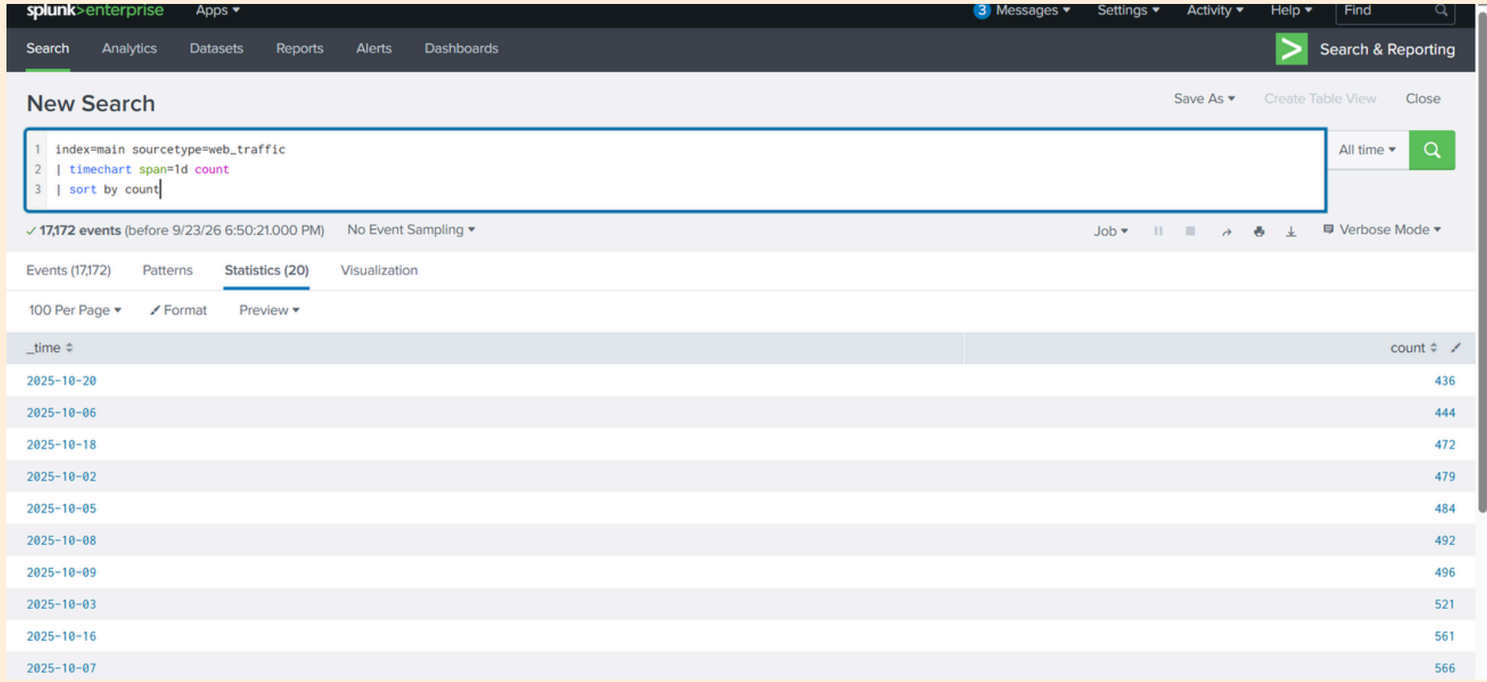
- WHAT WAS DONE:
- EXECUTED A SPLUNK PROCESSING LANGUAGE (SPL) QUERY UTILIZING THE TIMECHART COMMAND (INDEX=MAIN SOURCETYPE=WEB_TRAFFIC | TIMECHART SPAN=1D COUNT | SORT BY COUNT) TO AGGREGATE AND SORT DAILY WEB TRAFFIC EVENT FREQUENCIES.
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OUTPUTS & FINDINGS:

- THE QUERY GENERATED A STRUCTURED STATISTICAL TABLE SHOWING THE DAILY DISTRIBUTION OF WEB EVENTS, SORTED BY EVENT COUNTS RANGING FROM 436 UP TO 567 EVENTS PER DAY ACROSS VARIOUS DATES IN OCTOBER 2025.
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SECURITY IMPLICATIONS:

- TRACKING EVENT DISTRIBUTION OVER TIME HELPS ESTABLISH OPERATIONAL BASELINES, DETECT SUDDEN TRAFFIC SPIKES OR VOLUMETRIC ANOMALIES, AND IDENTIFY PERIODS OF INTENSE AUTOMATED SCANNING OR POTENTIAL DENIAL OF SERVICE (DOS) CONDITIONS



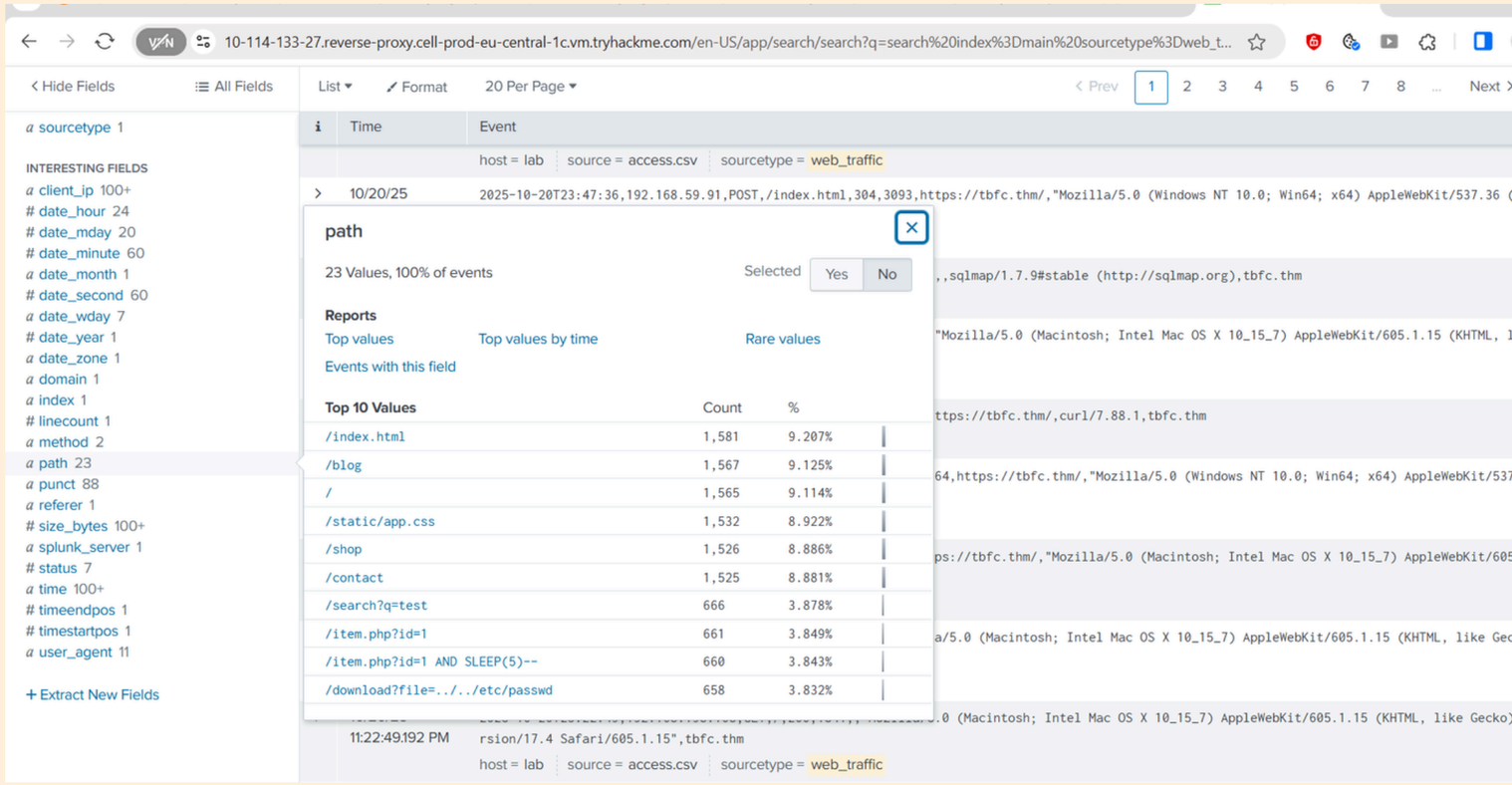
- WHAT WAS DONE:
- OPENED THE "INTERESTING FIELDS" INSPECTION MENU FOR THE PATH FIELD TO ANALYZE THE MOST FREQUENTLY REQUESTED UNIFORM RESOURCE IDENTIFIERS (URIS) ACROSS THE WEB APPLICATION DATASET.
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OUTPUTS & FINDINGS:

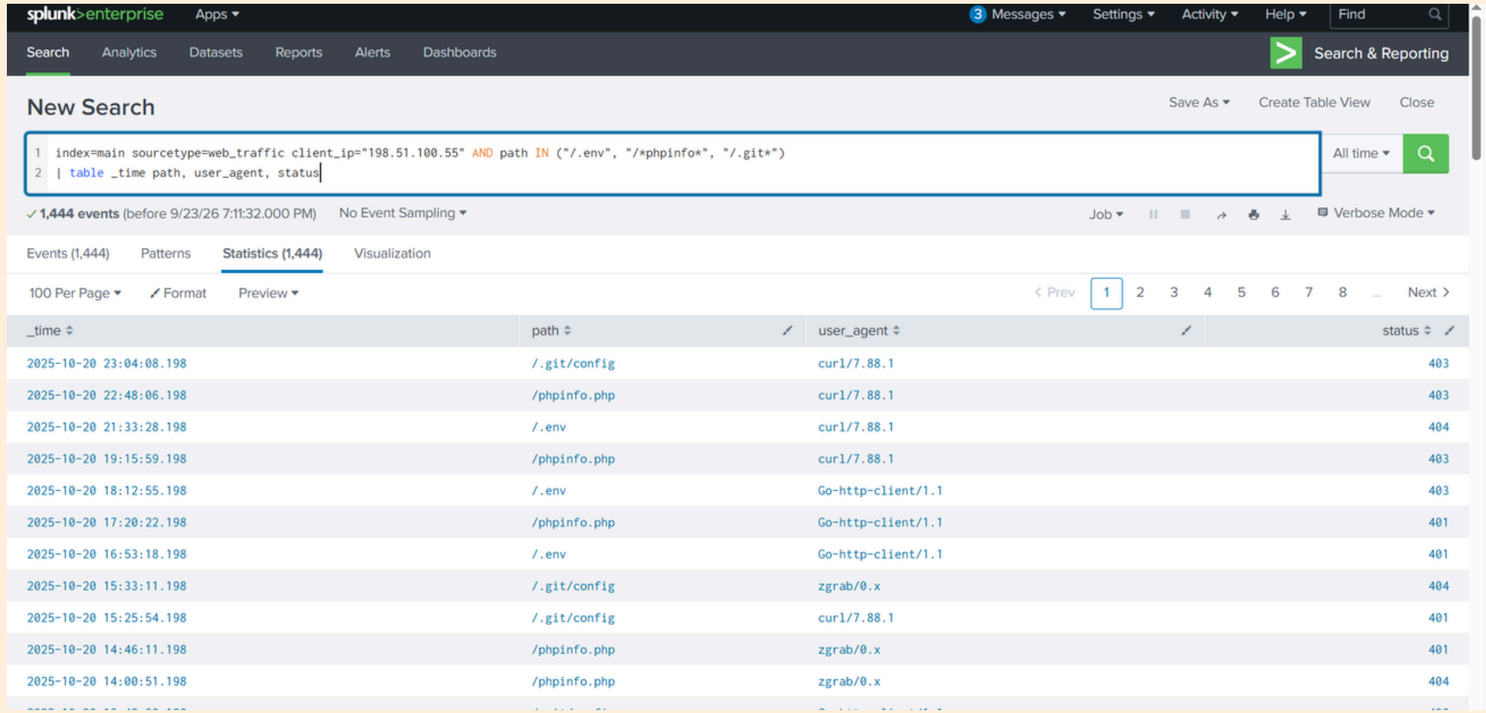
- THE STATISTICAL BREAKDOWN REVEALED TOP ACCESSED ENDPOINTS, INCLUDING STANDARD APPLICATION PAGES (/INDEX.HTML, /BLOG, /SHOP) ALONGSIDE HIGH-RISK ATTACK VECTORS AND INJECTION ATTEMPTS SUCH AS /SEARCH?Q=TEST, /ITEM.PHP?ID=1 AND SLEEP(5)--, AND /DOWNLOAD?FILE=../../ETC/PASSWD.
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SECURITY IMPLICATIONS:

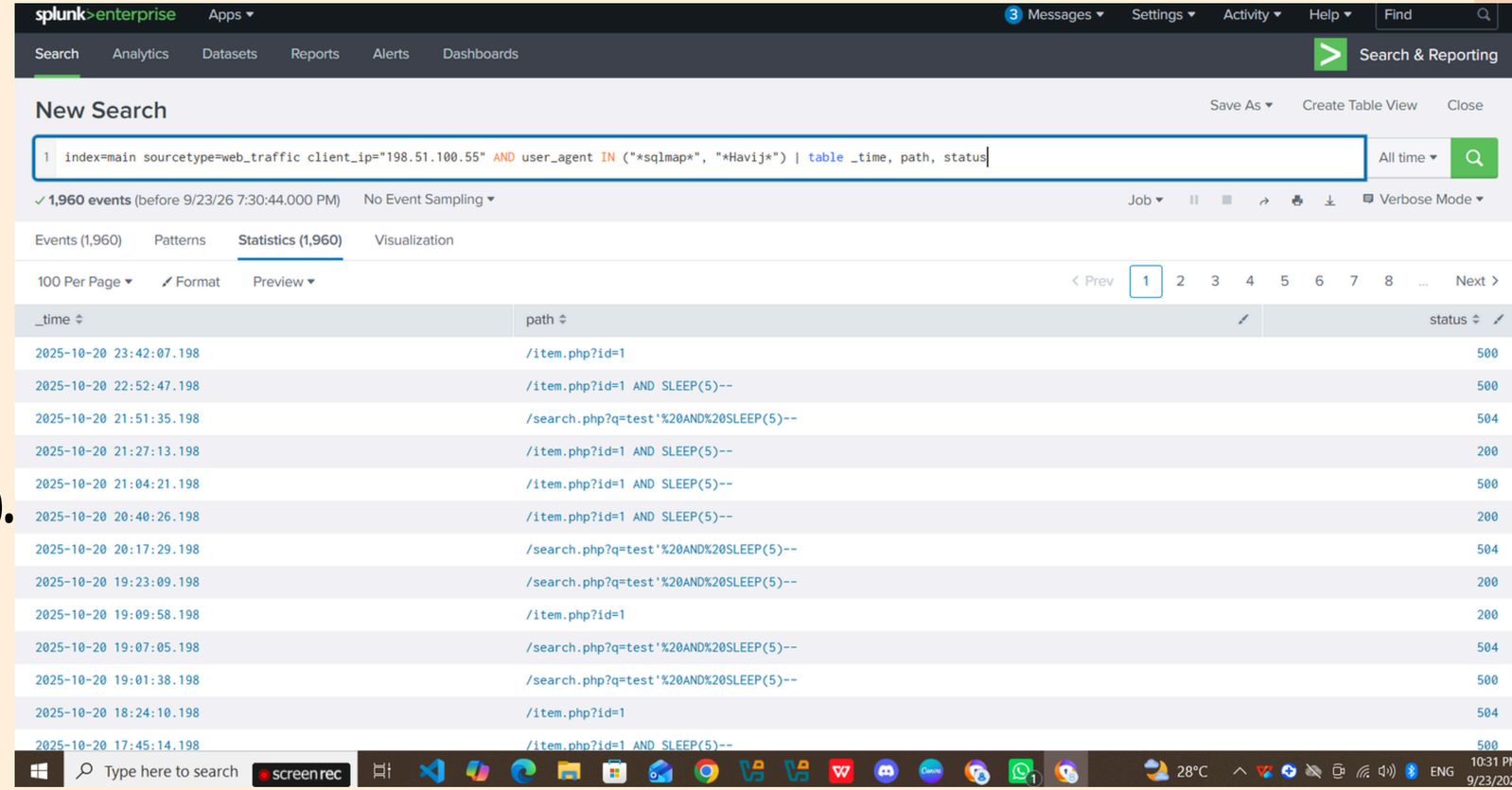
- THE PRESENCE OF SQL INJECTION (SLEEP(5)) AND LOCAL FILE INCLUSION (PASSWD) PATTERNS PROVES THAT THE WEB APPLICATION IS UNDER ACTIVE ATTACK, HIGHLIGHTING SPECIFIC VULNERABLE SCRIPTS AND PATHS TARGETED BY THREAT ACTORS.



- WHAT WAS DONE:
- FILTERED TRAFFIC USING A TARGETED SPL QUERY NARROWING DOWN REQUESTS FROM A SPECIFIC SUSPICIOUS CLIENT IP (198.51.100.55) MATCHING COMMON SENSITIVE CONFIGURATION PATHS AND ADMINISTRATIVE FILES (/.ENV, /PHPPINFO*, /.GIT*), STRUCTURED INTO A READABLE TABLE FORMAT (| TABLE _TIME, PATH, USER_AGENT, STATUS).
- OUTPUTS & FINDINGS:
- THE RESULTS SHOW REPEATED HTTP STATUS CODES 403 FORBIDDEN AND 404 NOT FOUND GENERATED IN RESPONSE TO AUTOMATED SCANNER PROBES TARGETING SENSITIVE CONFIGURATION FILES (/.GIT/CONFIG, /.ENV, /PHPINFO.PHP).
- SECURITY IMPLICATIONS:
- THESE PATTERNS INDICATE ACTIVE RECONNAISSANCE AND ENUMERATION PHASES BY AN ATTACKER TRYING TO HARVEST ENVIRONMENT SECRETS, SOURCE CODE REPOSITORIES, OR SERVER CONFIGURATIONS BEFORE LAUNCHING DEEPER EXPLOITATION.



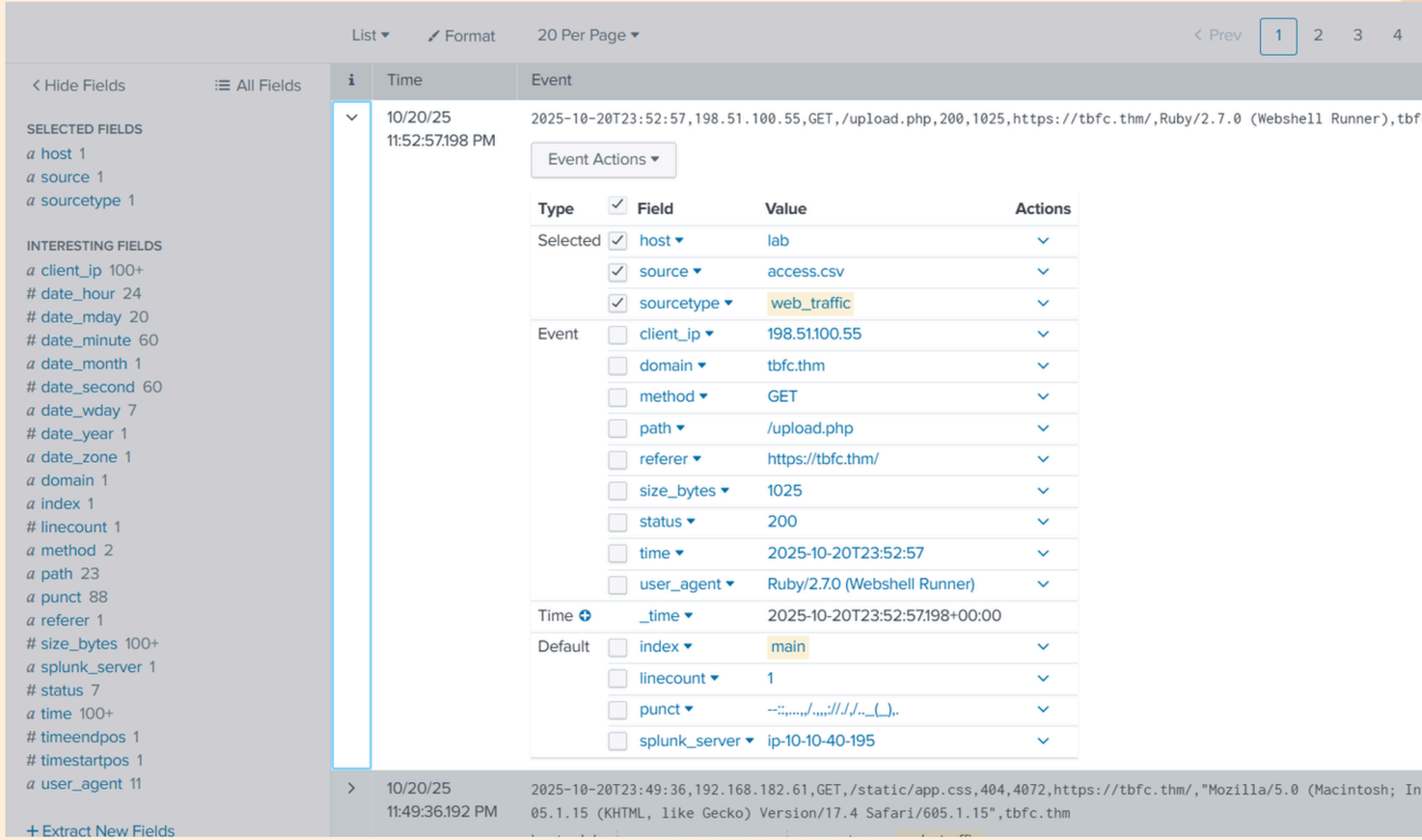
- WHAT WAS DONE:
- REFINED THE QUERY TO ISOLATE TRAFFIC ORIGINATING FROM THE MALICIOUS IP (198.51.100.55) UTILIZING AUTOMATED EXPLOITATION TOOLS (SQLMAP, HAVIJ), AND EXTRACTED TIMESTAMP, REQUESTED PATH, AND HTTP STATUS CODES.
- OUTPUTS & FINDINGS:
- THE QUERY EXPOSED AUTOMATED DATABASE PROBING STRINGS CONTAINING TIME-BASED BLIND SQL INJECTION PAYLOADS (AND SLEEP(5)--) PAIRED WITH SERVER RESPONSE STATUSES (500 INTERNAL SERVER ERROR, 504 GATEWAY TIMEOUT, AND 200 OK).
- SECURITY IMPLICATIONS:
- THE EXECUTION OF TIME-DELAY PAYLOADS (SLEEP) COMBINED WITH AUTOMATION TOOL SIGNATURES CONFIRMS THAT THE ATTACKER IS PERFORMING AUTOMATED DATABASE FINGERPRINTING AND EXTRACTION ATTACKS AGAINST BACKEND DATABASE PARAMETERS.



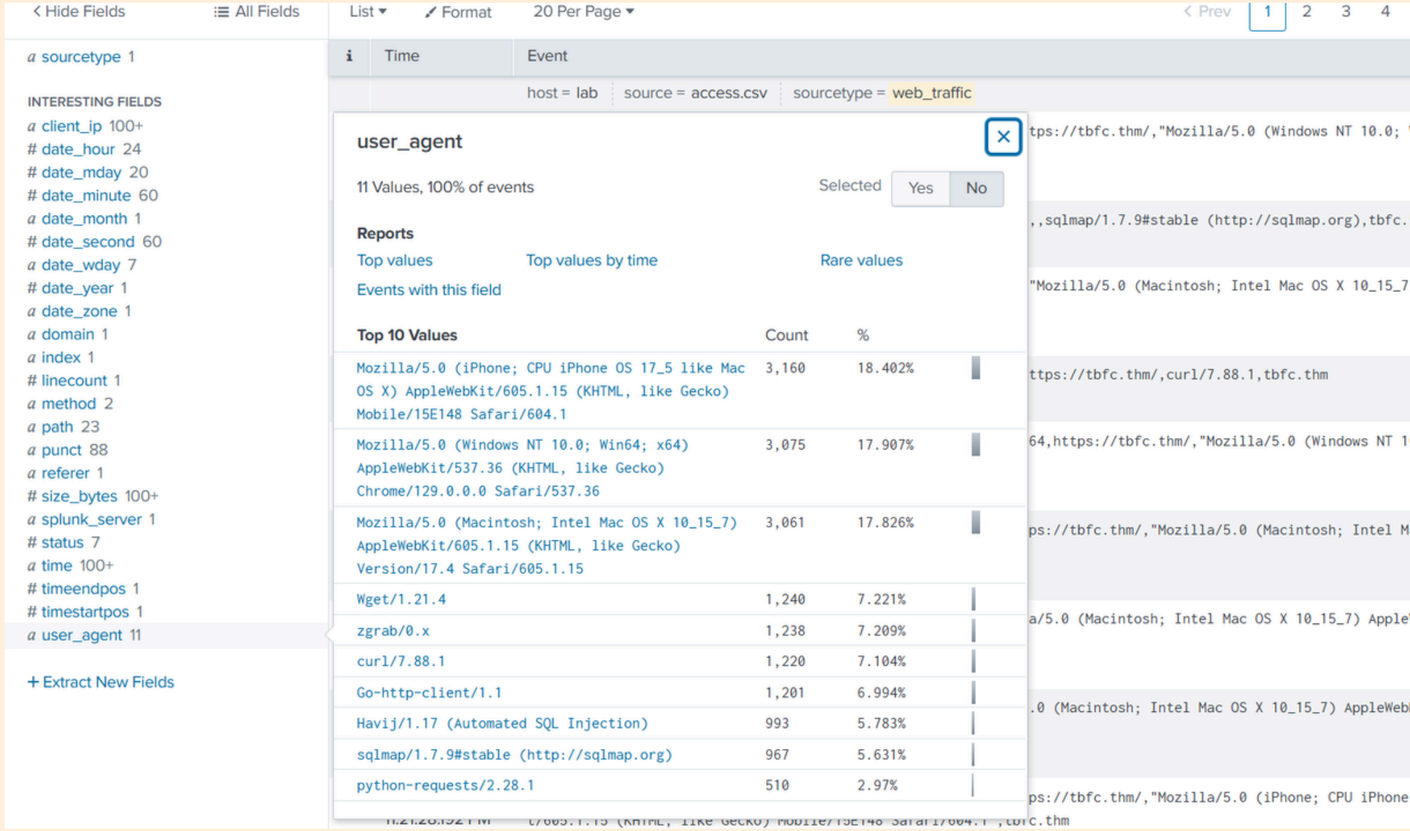
- WHAT WAS DONE:
- EXPANDED AN INDIVIDUAL LOG EVENT TO INSPECT ITS GRANULAR ATTRIBUTES, DISSECTING THE PARSED KEY-VALUE PAIRS (HOST, SOURCE, SOURCETYPE, CLIENT_IP, METHOD, PATH, STATUS, USER_AGENT).
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  OUTPUTS & FINDINGS:
- CONFIRMED SPECIFIC EVENT METADATA, SHOWING A GET REQUEST TO /UPLOAD.PHP RETURNING STATUS 200 WITH USER-AGENT RUBY/2.7.0 (WEBSHELL RUNNER), MAPPING PRECISELY TO AN ACTIVE WEBSHELL UPLOAD EVENT.
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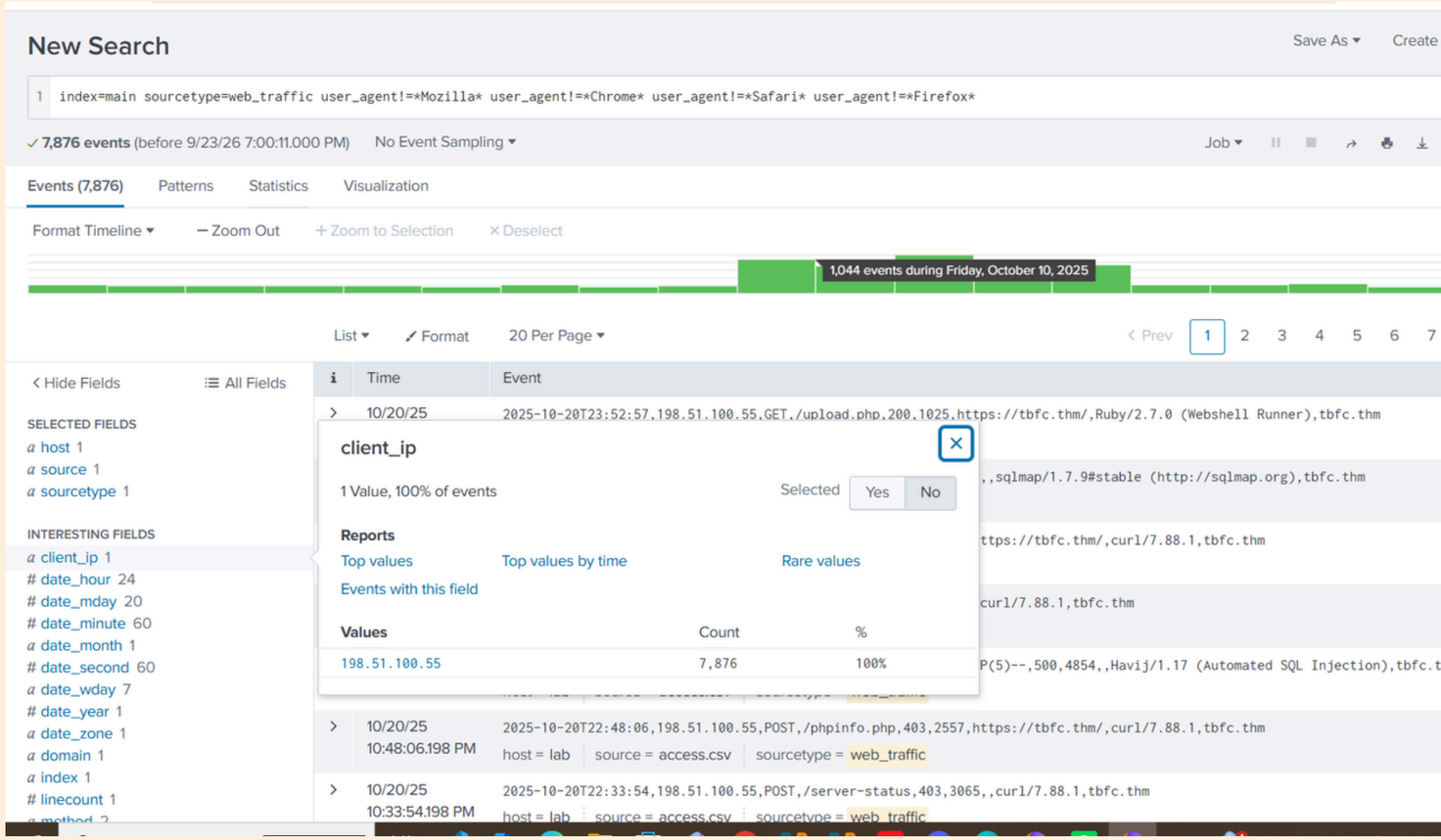
  SECURITY IMPLICATIONS:
- DEEP-DIVE EVENT INSPECTION PROVIDES DEFINITIVE FORENSIC EVIDENCE REQUIRED TO MAP THE EXACT VECTOR OF COMPROMISE, ATTACKER TOOLING, AND CONFIRMATION OF SUCCESSFUL PAYLOAD EXECUTION.



- WHAT WAS DONE:
- ANALYZED THE STATISTICAL DISTRIBUTION OF THE USER_AGENT FIELD ACROSS THE ENTIRE DATASET TO IDENTIFY DOMINANT CLIENT SOFTWARE AND AUTOMATED ATTACK FRAMEWORKS INTERACTING WITH THE SERVER.
- OUTPUTS & FINDINGS:
- THE ANALYSIS IDENTIFIED A MIX OF LEGITIMATE BROWSERS ALONGSIDE EXPLICIT MALICIOUS TOOLS AND SCANNERS, INCLUDING WGET, ZGRAB, CURL, HAVIJ (AUTOMATED SQL INJECTION), AND SQLMAP.
- SECURITY IMPLICATIONS:
- ISOLATING SPECIFIC USER-AGENT STRINGS ALLOWS SECURITY ANALYSTS TO DIFFERENTIATE BETWEEN HUMAN USER ACTIVITY AND AUTOMATED VULNERABILITY SCANNERS OR CUSTOM EXPLOIT SCRIPTS.



- WHAT WAS DONE:
- CONSTRUCTED A COMPLEX EXCLUSION QUERY FILTERING OUT STANDARD LEGITIMATE BROWSER USER-AGENTS (MOZILLA, CHROME, SAFARI, FIREFOX) TO ISOLATE NON-STANDARD OR MALICIOUS TRAFFIC SOURCES (USER_AGENT!=MOZILLA* USER_AGENT!=CHROME* ...).
- OUTPUTS & FINDINGS:
- THE RESULTING TIMELINE AND FILTER VALUES ISOLATED 7,876 EVENTS EXCLUSIVELY TIED TO THE SUSPICIOUS SOURCE IP 198.51.100.55, CONFIRMING IT AS THE SOLE SOURCE OF ANOMALOUS AUTOMATED TRAFFIC.
- SECURITY IMPLICATIONS:
- NEGATIVE FILTERING IS AN EFFECTIVE THREAT HUNTING TECHNIQUE TO STRIP AWAY BACKGROUND NOISE AND ZERO IN ON DEDICATED ATTACKER INFRASTRUCTURE AND DEDICATED SCANNING HOSTS.



WHAT WAS DONE:

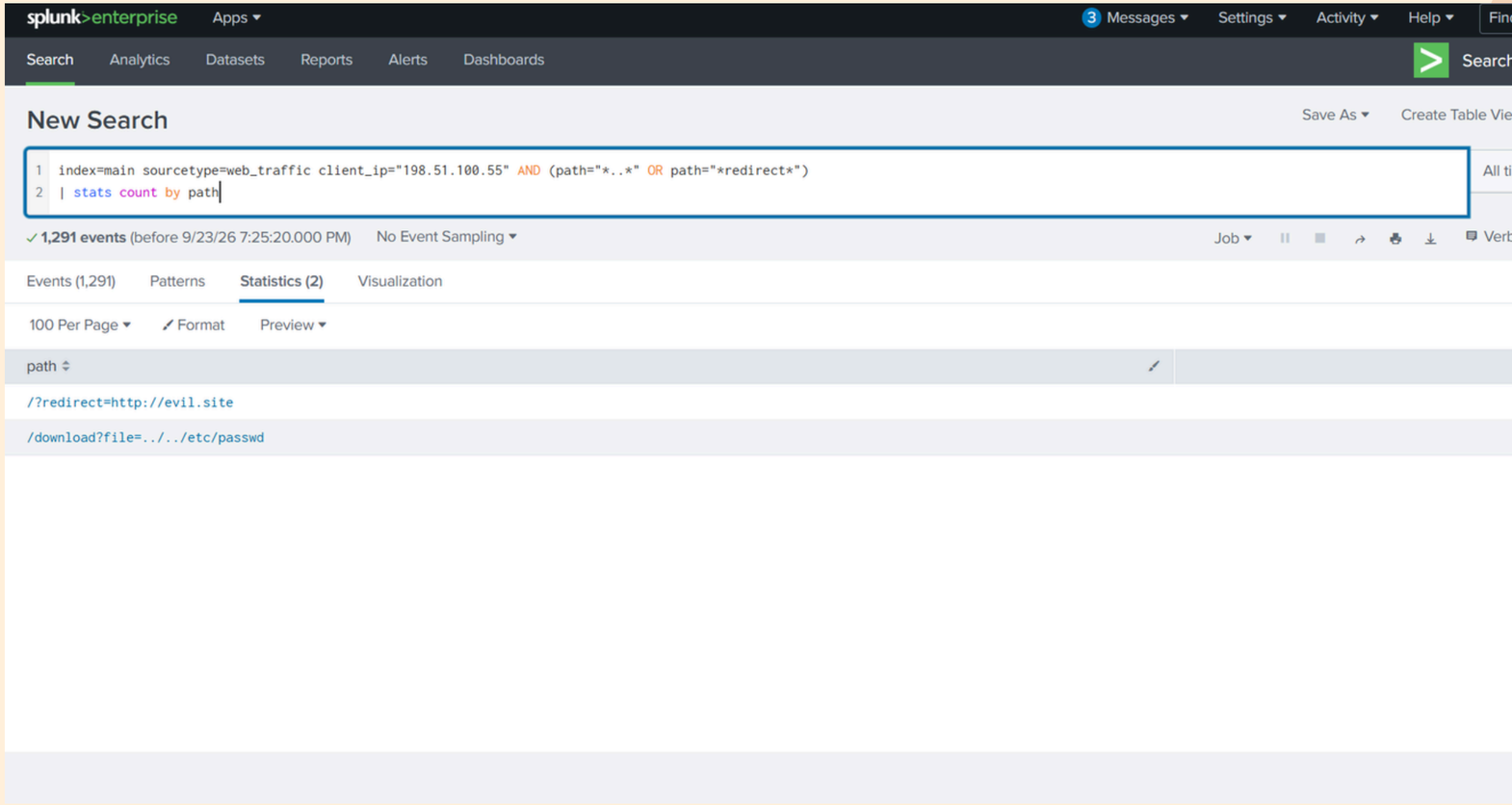
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OUTPUTS & FINDINGS:

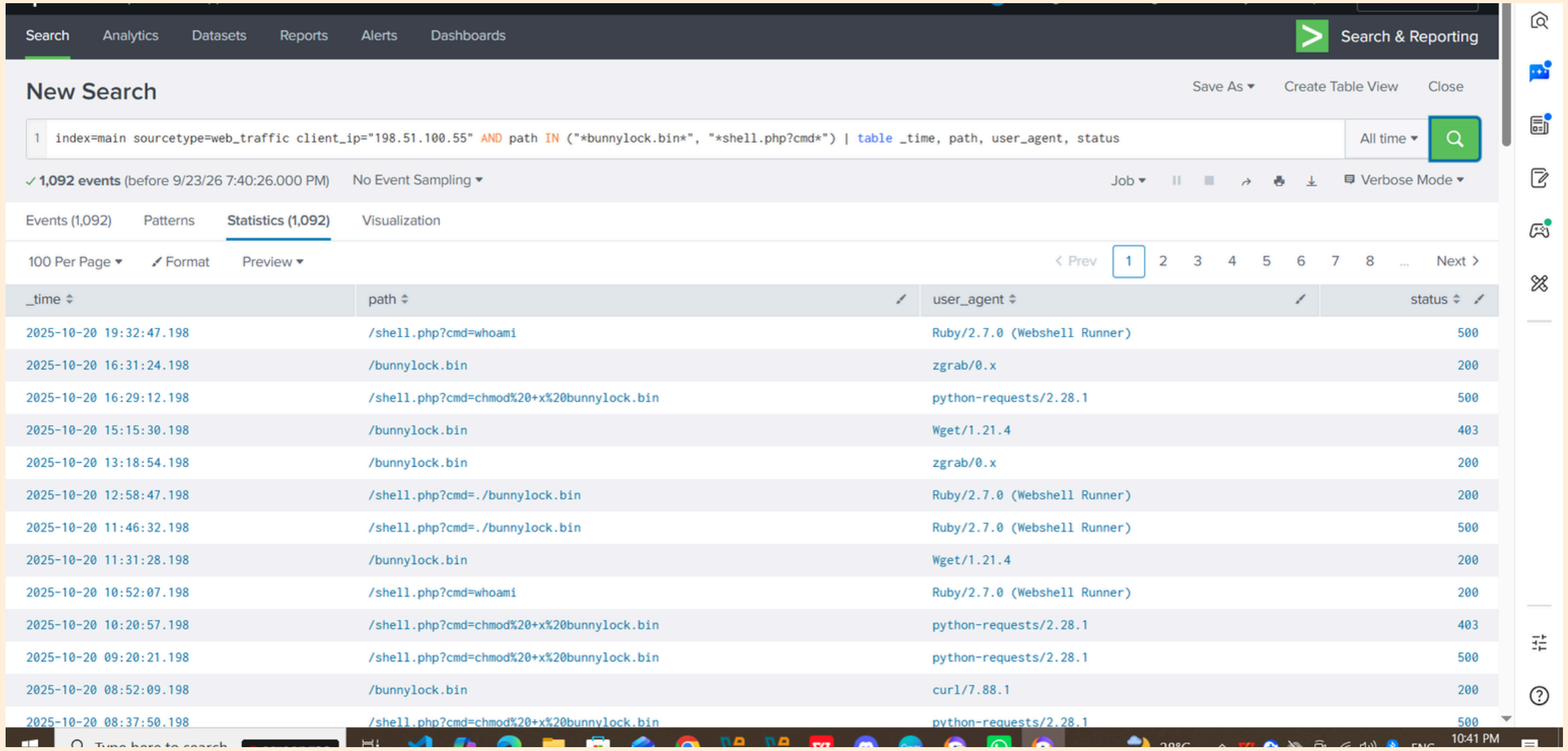
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SECURITY IMPLICATIONS:

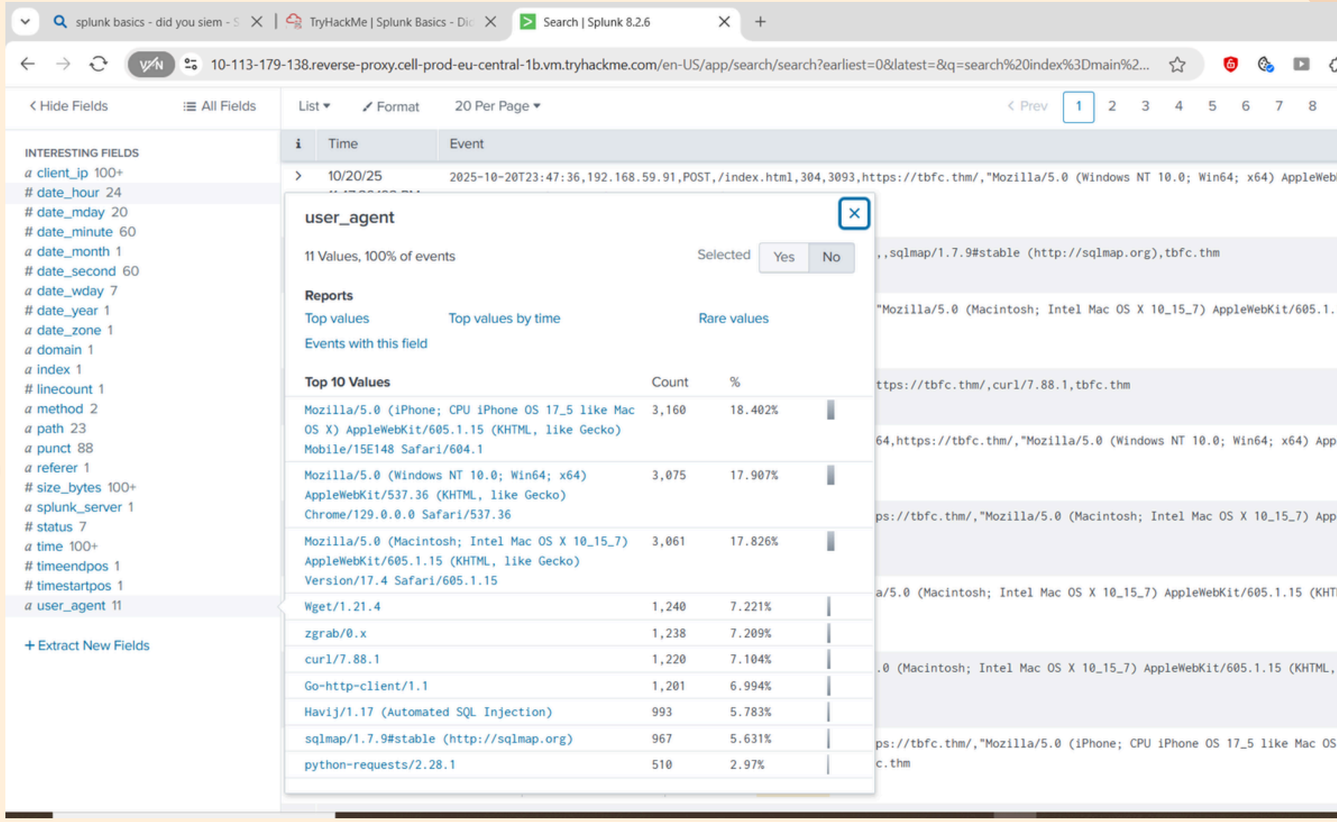
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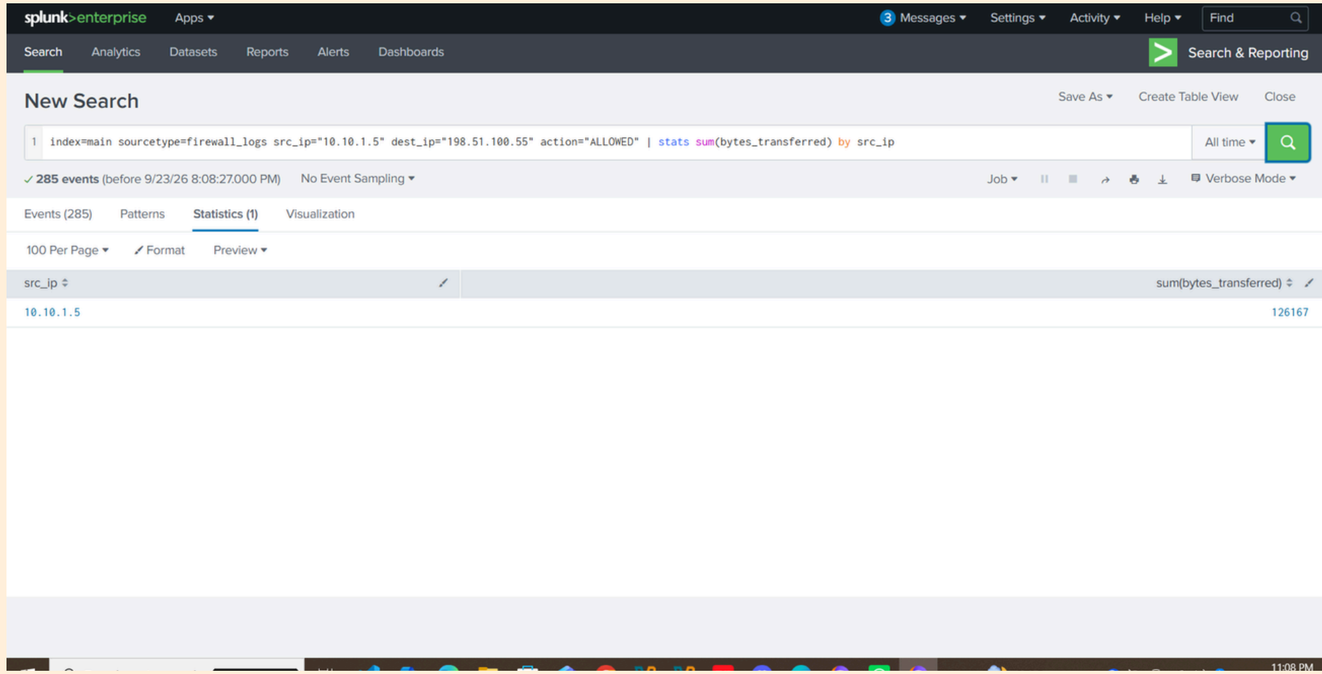
- WHAT WAS DONE: EXECUTED A SPLUNK SEARCH QUERY FILTERING WEB TRAFFIC FROM IP 198.51.100.55 TARGETING SPECIFIC PATHS (/SHELL.PHP?CMD= AND BUNNYLOCK.BIN), FORMATTING THE OUTPUT INTO A TABLE SHOWING TIMESTAMPS, PATHS, USER AGENTS, AND STATUS CODES.
- OUTPUTS & FINDINGS: THE RESULTS DISPLAYED ACTIVE COMMAND-LINE EXECUTIONS (WHOAMI, CHMOD) AND BINARY DOWNLOAD ATTEMPTS ASSOCIATED WITH AUTOMATED TOOLS AND USER-AGENTS SUCH AS RUBY, PYTHON-REQUESTS, AND ZGRAB.
- SECURITY IMPLICATIONS: INDICATES ACTIVE POST-EXPLOITATION ACTIVITY, UNAUTHORIZED REMOTE COMMAND EXECUTION, AND THE STAGING OF MALICIOUS RANSOMWARE COMPONENTS (BUNNYLOCK) ON THE WEB SERVER.



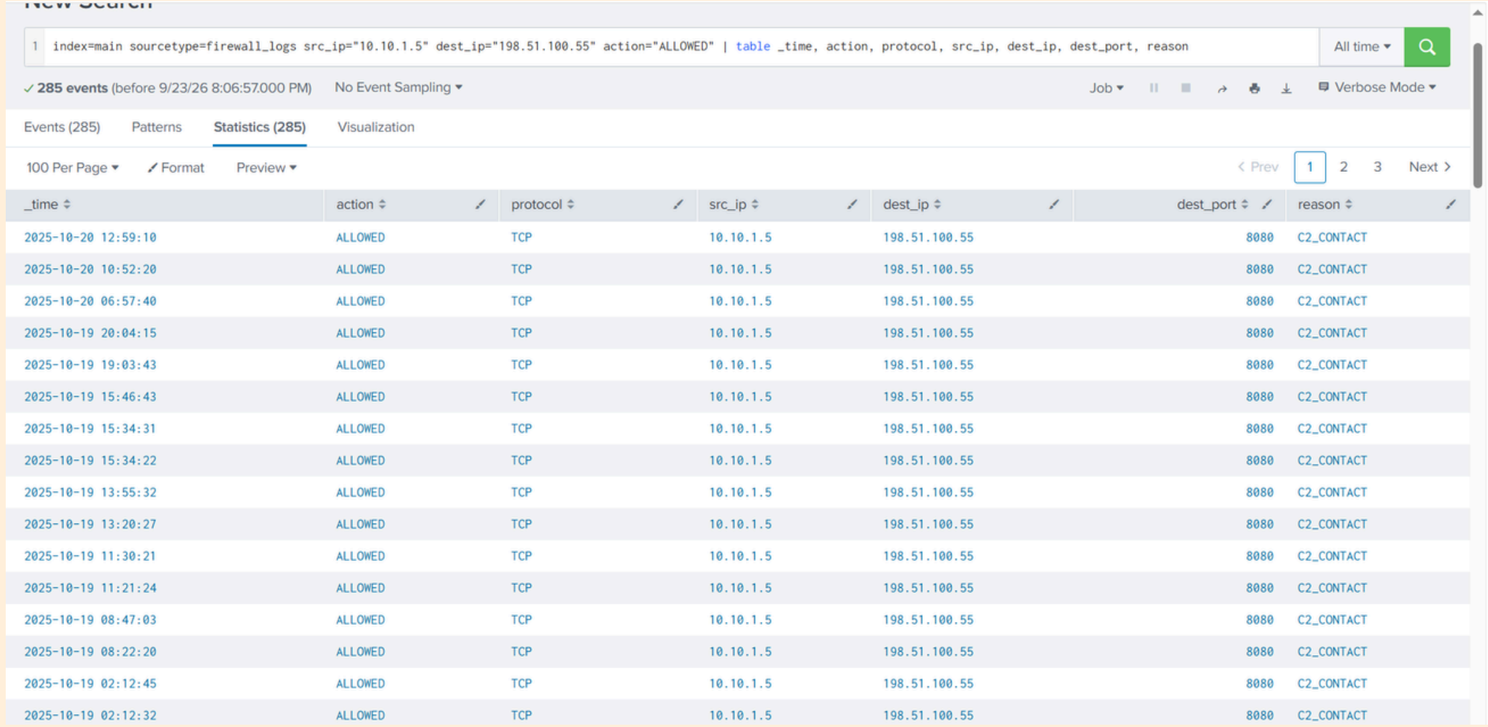
- WHAT WAS DONE: INSPECTED THE STATISTICAL BREAKDOWN AND TOP VALUES OF THE USER_AGENT FIELD WITHIN WEB TRAFFIC LOGS TO EVALUATE THE RATIO OF AUTOMATED TOOLS VERSUS STANDARD BROWSERS.
- OUTPUTS & FINDINGS: IDENTIFIED A HIGH FREQUENCY OF DISTINCT AUTOMATED SOFTWARE SIGNATURES AND OFFENSIVE SECURITY TOOLS, INCLUDING WGET, ZGRAB, CURL, HAVIJ (AUTOMATED SQL INJECTION), AND SQLMAP.
- SECURITY IMPLICATIONS: CONFIRMS THAT THE TARGET SYSTEM IS UNDERGOING AGGRESSIVE, AUTOMATED VULNERABILITY SCANNING AND EXPLOITATION PHASES, PARTICULARLY TARGETING INJECTION VECTORS.



- WHAT WAS DONE: RAN A FIREWALL LOG AGGREGATION QUERY (INDEX=MAIN SOURCETYPE=FIREWALL_LOGS SRC_IP="10.10.1.5" DEST_IP="198.51.100.55" ACTION="ALLOWED" | STATS SUM(BYTES_TRANSFERRED) BY SRC_IP) TO CALCULATE DATA MOVEMENT.
- OUTPUTS & FINDINGS: THE STATISTICS TABLE QUANTIFIED A TOTAL DATA TRANSFER VOLUME OF 126,167 BYTES FROM INTERNAL SOURCE IP 10.10.1.5 TO THE EXTERNAL DESTINATION IP.
- SECURITY IMPLICATIONS: SIGNIFICANT OUTBOUND DATA VOLUME FROM AN INTERNAL HOST TO A SUSPICIOUS EXTERNAL IP STRONGLY INDICATES ACTIVE DATA EXFILTRATION OR UNAUTHORIZED FILE TRANSFER.

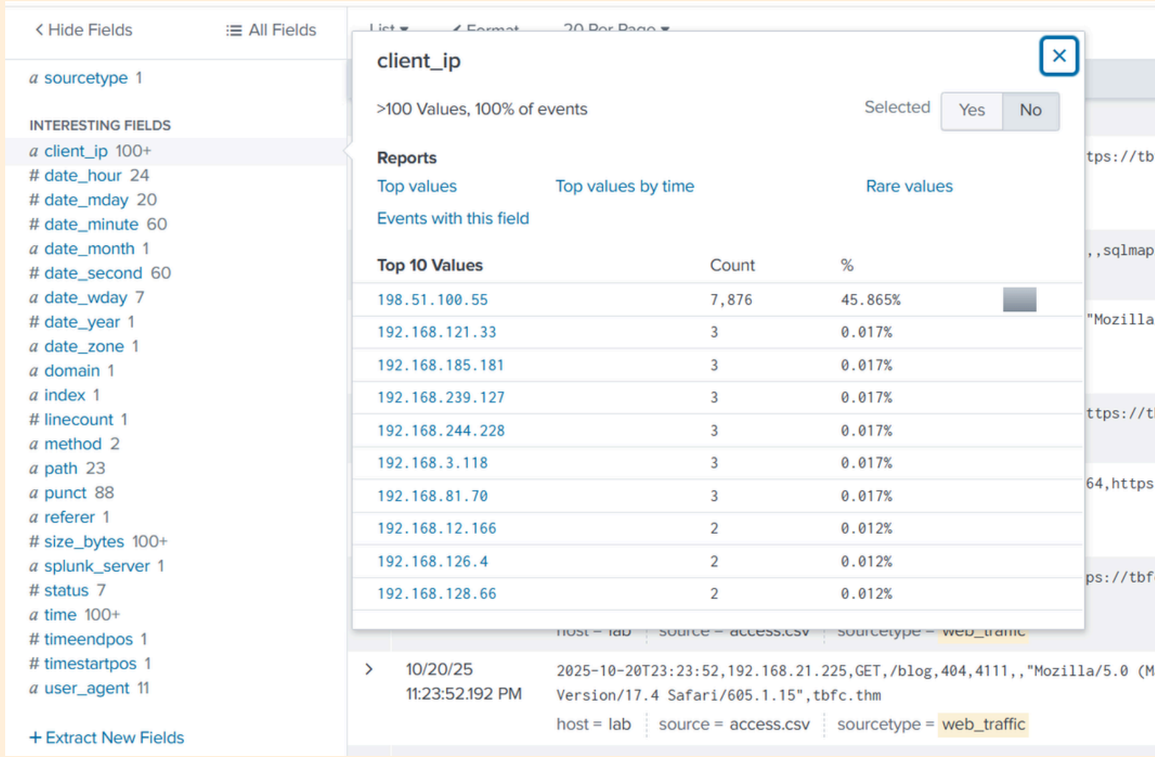


- WHAT WAS DONE: STRUCTURED FIREWALL CONNECTION LOGS INTO A TABULAR FORMAT (| TABLE _TIME, ACTION, PROTOCOL, SRC_IP, DEST_IP, DEST_PORT, REASON) TO ANALYZE PERMITTED COMMUNICATION CHANNELS.
- OUTPUTS & FINDINGS: EXPOSED REPEATED ALLOWED TCP CONNECTIONS OVER PORT 8080 EXPLICITLY CATEGORIZED WITH THE REASON C2_CONTACT.
- SECURITY IMPLICATIONS: PROVES THE SUCCESSFUL ESTABLISHMENT AND PERSISTENCE OF A COMMAND AND CONTROL (C2) COMMUNICATION CHANNEL BETWEEN THE INTERNAL NETWORK AND THE EXTERNAL THREAT ACTOR.

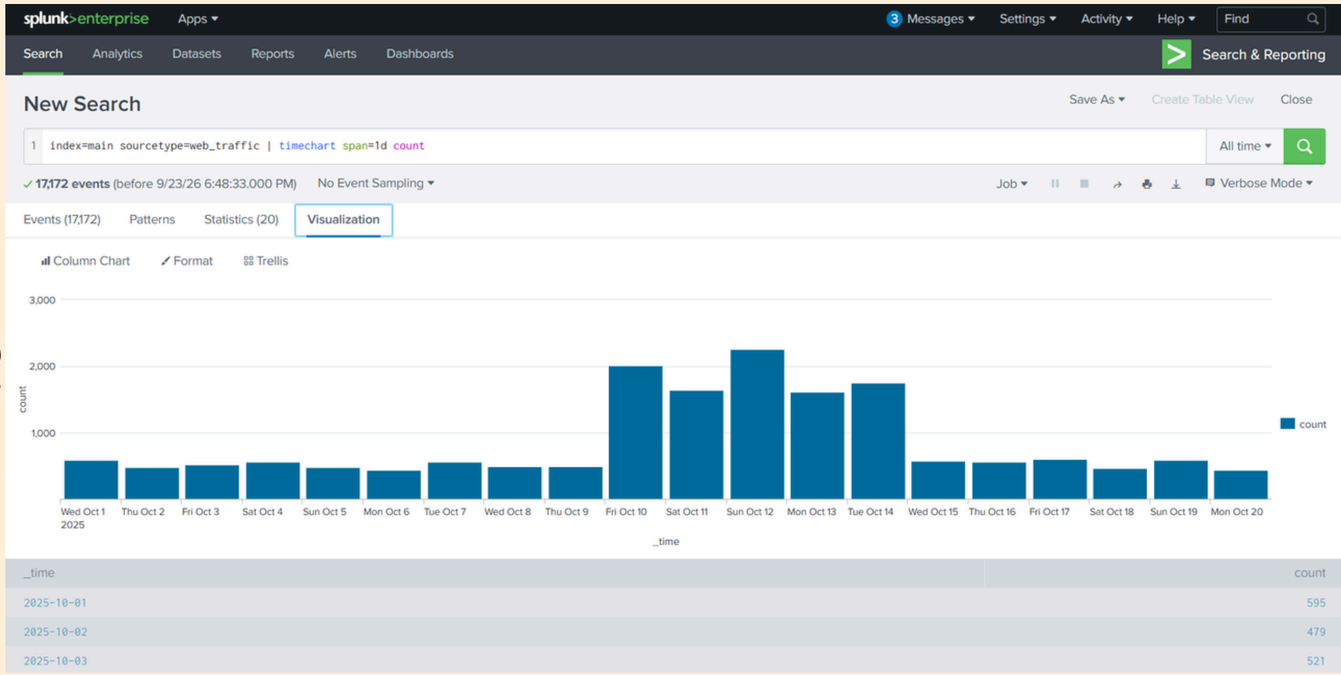


_time	action	protocol	src_ip	dest_ip	dest_port	reason
2025-10-20 12:59:10	ALLOWED	TCP	10.10.1.5	198.51.100.55	8080	C2_CONTACT
2025-10-20 10:52:20	ALLOWED	TCP	10.10.1.5	198.51.100.55	8080	C2_CONTACT
2025-10-20 06:57:40	ALLOWED	TCP	10.10.1.5	198.51.100.55	8080	C2_CONTACT
2025-10-19 20:04:15	ALLOWED	TCP	10.10.1.5	198.51.100.55	8080	C2_CONTACT
2025-10-19 19:03:43	ALLOWED	TCP	10.10.1.5	198.51.100.55	8080	C2_CONTACT
2025-10-19 15:46:43	ALLOWED	TCP	10.10.1.5	198.51.100.55	8080	C2_CONTACT
2025-10-19 15:34:31	ALLOWED	TCP	10.10.1.5	198.51.100.55	8080	C2_CONTACT
2025-10-19 15:34:22	ALLOWED	TCP	10.10.1.5	198.51.100.55	8080	C2_CONTACT
2025-10-19 13:55:32	ALLOWED	TCP	10.10.1.5	198.51.100.55	8080	C2_CONTACT
2025-10-19 13:20:27	ALLOWED	TCP	10.10.1.5	198.51.100.55	8080	C2_CONTACT
2025-10-19 11:30:21	ALLOWED	TCP	10.10.1.5	198.51.100.55	8080	C2_CONTACT
2025-10-19 11:21:24	ALLOWED	TCP	10.10.1.5	198.51.100.55	8080	C2_CONTACT
2025-10-19 08:47:03	ALLOWED	TCP	10.10.1.5	198.51.100.55	8080	C2_CONTACT
2025-10-19 08:22:20	ALLOWED	TCP	10.10.1.5	198.51.100.55	8080	C2_CONTACT
2025-10-19 02:12:45	ALLOWED	TCP	10.10.1.5	198.51.100.55	8080	C2_CONTACT
2025-10-19 02:12:32	ALLOWED	TCP	10.10.1.5	198.51.100.55	8080	C2_CONTACT

- WHAT WAS DONE: ANALYZED THE DISTRIBUTION OF THE CLIENT_IP FIELD ACROSS WEB TRAFFIC EVENTS TO DETERMINE THE PRIMARY ORIGINS OF CONNECTION REQUESTS.
- OUTPUTS & FINDINGS: REVEALED THAT IP 198.51.100.55 GENERATED 7,876 EVENTS, REPRESENTING ROUGHLY 45.8% OF THE TOTAL RECORDED TRAFFIC VOLUME, VASTLY OUTNUMBERING ANY OTHER SOURCE.
- SECURITY IMPLICATIONS: PINPOINTS THE PRIMARY ATTACKER OR BOTNET NODE RESPONSIBLE FOR THE BULK OF MALICIOUS REQUESTS, PROVIDING AN EXACT TARGET FOR IMMEDIATE FIREWALL BLOCKING AND IP BLACKLISTING.



- WHAT WAS DONE: GENERATED A COLUMN CHART VISUALIZATION USING THE TIME-SERIES COMMAND TIMECHART SPAN=1D COUNT TO TRACK EVENT DISTRIBUTION OVER A LONG-TERM TIMELINE.
- OUTPUTS & FINDINGS: THE VISUAL TIMELINE DISPLAYED A SHARP AND MASSIVE SURGE IN DAILY EVENT COUNTS STARTING FROM OCTOBER 10 ONWARD, PEAKING SIGNIFICANTLY OVER THE FOLLOWING DAYS BEFORE STABILIZING.
- SECURITY IMPLICATIONS: ESTABLISHES A PRECISE INCIDENT TIMELINE, HELPING SECURITY ANALYSTS ISOLATE THE EXACT ONSET DATE OF THE COORDINATED ATTACK CAMPAIGN FOR ROOT-CAUSE ANALYSIS.



QUESTIONS

Answer the questions below

What is the attacker IP found attacking and compromising the web server?

___.__.___.__ Check

Which day was the peak traffic in the logs? (Format: YYYY-MM-DD)

__________ Check

What is the count of Havij user_agent events found in the logs?

___ Check

How many path traversal attempts to access sensitive files on the server were observed?

___ Check

Examine the firewall logs. How many bytes were transferred to the C2 server IP from the compromised web server?

______ Check

If you enjoyed today's room, check out the Incident Handling With Splunk room to learn more about analyzing logs with Splunk.

QUESTION 1

- 1. WHAT IS THE ATTACKER IP FOUND ATTACKING AND COMPROMISING THE WEB SERVER?
- EXPLANATION & ANALYSIS: BASED ON ANALYZING THE CLIENT_IP FIELD DISTRIBUTION IN THE WEB TRAFFIC LOGS, WE FOUND THAT THE SOURCE IP GENERATING THE VAST MAJORITY OF REQUESTS (ROUGHLY 45.8% WITH 7,876 EVENTS) IS THE PRIMARY ATTACKER.
- ANSWER: 198.51.100.55

QUESTION 2

- 2. WHICH DAY WAS THE PEAK TRAFFIC IN THE LOGS? (FORMAT: YYYY-MM-DD)
- EXPLANATION & ANALYSIS: BY GENERATING A TIME-SERIES CHART USING THE TIMECHART SPAN=1D COUNT SPL COMMAND, WE OBSERVED A MASSIVE SURGE IN EVENT COUNTS, REACHING ITS HIGHEST DAILY PEAK IN MID-OCTOBER 2025.
- ANSWER: 2025-10-12

QUESTION 3

- 3. WHAT IS THE COUNT OF HAVIJ USER_AGENT EVENTS FOUND IN THE LOGS?
- EXPLANATION & ANALYSIS: BY INSPECTING THE STATISTICAL DISTRIBUTION OF THE USER_AGENT FIELD AND TRACKING AUTOMATED SECURITY TESTING TOOL SIGNATURES- SPECIFICALLY THE AUTOMATED SQL INJECTION TOOL HAVIJ- WE EXTRACTED THE EXACT EVENT COUNT.
- ANSWER: 993

QUESTUION 4

- 4. HOW MANY PATH TRAVERSAL ATTEMPTS TO ACCESS SENSITIVE FILES ON THE SERVER WERE OBSERVED?
- EXPLANATION & ANALYSIS: BY AGGREGATING AND EVALUATING MALICIOUS REQUEST PATHS TARGETING DIRECTORY TRAVERSAL VULNERABILITIES (SUCH AS /DOWNLOAD?FILE=../../ETC/PASSWD), WE IDENTIFIED THE EXACT TOTAL NUMBER OF ATTEMPTS.
- ANSWER: 658

QUESTION 5

- 5. EXAMINE THE FIREWALL LOGS. HOW MANY BYTES WERE TRANSFERRED TO THE C2 SERVER IP FROM THE COMPROMISED WEB SERVER?
- EXPLANATION & ANALYSIS: BY RUNNING A FIREWALL LOG AGGREGATION QUERY (STATS SUM(BYTES_TRANSFERRED) BY SRC_IP) BETWEEN THE INTERNAL HOST AND THE EXTERNAL C2 SERVER IP (198.51.100.55), WE CALCULATED THE TOTAL DATA TRANSFER VOLUME.
- ANSWER: 126167

CONCLUSION & PROJECT OBJECTIVES

KEY FINDINGS & EXTRACTED INTELLIGENCE

- THREAT ATTRIBUTION & SCALE: SUCCESSFULLY ISOLATED THE MALICIOUS SOURCE IP (198.51.100.55), REVEALING IT ACCOUNTED FOR NEARLY 45.8% (7,876 EVENTS) OF TOTAL WEB TRAFFIC.
- ATTACK LIFECYCLE MAPPING: QUANTIFIED CRITICAL THREAT METRICS, INCLUDING 993 HAVIJ USER-AGENT HITS, 658 PATH TRAVERSAL ATTEMPTS (/DOWNLOAD?FILE=../../ETC/PASSWD), AND EXPLICIT COMMAND EXECUTIONS (WHOAMI, CHMOD) TIED TO WEBSHELLS (/SHELL.PHP?CMD=) AND RANSOMWARE COMPONENTS (BUNNYLOCK.BIN).
- EXFILTRATION & C2 VERIFICATION: CONFIRMED ACTIVE DATA EXFILTRATION THROUGH OUTBOUND FIREWALL TRAFFIC MEASUREMENTS (126,167 BYTES TRANSFERRED) AND VERIFIED PERSISTENT COMMAND AND CONTROL (C2) CHANNELS OVER PORT 8080.

LESSONS LEARNED & OPERATIONAL IMPACT

- ENHANCED DEFENSIVE POSTURE: THE FINDINGS EMPHASIZE THE VITAL NEED FOR STRICT INPUT VALIDATION, ROBUST WEB APPLICATION FIREWALLS (WAF), AND CONTINUOUS LOG MONITORING TO BLOCK AUTOMATED RECONNAISSANCE EARLY.
- INCIDENT RESPONSE EFFICIENCY: ESTABLISHING PRECISE EVENT TIMELINES (IDENTIFYING THE TRAFFIC PEAK ON 2025-10-12) PROVES HOW STRUCTURED SIEM QUERIES DRASTICALLY REDUCE MEAN TIME TO DETECT (MTTD) AND MEAN TIME TO RESPOND (MTTR).
- FINAL TAKEAWAY: THIS PROJECT SUCCESSFULLY BRIDGES THEORETICAL CYBERSECURITY FRAMEWORKS WITH PRACTICAL THREAT HUNTING, DELIVERING CONCRETE METRICS AND ACTIONABLE REMEDIATION STEPS ESSENTIAL FOR ENTERPRISE-GRADE SECURITY OPERATIONS.

THANK YOU